Errata and Counterexamples for *Approximate Majority Analyses using Tri-molecular Chemical Reaction Networks*

Condon, Hajiaghayi, Kirkpatrick, Maňuch

found while formalizing the paper in Lean 4 / Mathlib

Xiang Huang

Abstract

These errata were found while producing a machine-checked formalization of the paper in Lean 4 with Mathlib. Each entry identifies the exact printed passage, gives either a short quotation or an exact mathematical transcription, and states the correction; where a corrected theorem has been machine-checked we name the corresponding Lean declaration. Every location below was checked against the published Springer PDF of *Natural Computing* 19 (2020), 249–270, DOI `10.1007/s11047-019-09756-4`. We give both the printed publication page and the one-based PDF page, counted from the article's title page.

The paper's entry and consensus conclusions are not refuted by this audit. One additional persistence clause — Theorem 4's preservation of relaxed consensus for the following n^γ interactions — is false as printed. The remaining items are proof gaps, omitted hypotheses or qualifiers, and typographical errors. We do *not* claim that every downstream use survives: the false preservation clause has no corrected downstream proof at its printed strength. The full-range tail of Lemma 12 / Theorem 5 is true, but the printed proof does not establish it; a replacement state-dependent proof has been machine-checked.

Classification scheme. (E) a printed statement is false; (P) a genuine proof erratum with the statement still true; (U) an assumption, scope condition, probability qualifier, or stronger conclusion is under-stated; (T) a typo or typesetting omission with unambiguous intended meaning; and (O) *open after audit* — the printed proof does not establish the stated parameter range, and no counterexample or replacement proof is currently known.

The (O) label exists because E/P/U/T do not honestly cover a claim whose proof fails while its statement is neither proved nor refuted: labelling such a claim (P) would, by the definition above, assert that the statement is known true. Exactly one defect is (E) — Theorem 4's preservation clause, recorded first below and refuted by an explicit fixed-adversary family. No defect remains in class (O): the former open case, the full-range Lemma 12 / Theorem 5 tail, is now classified (P). The replacement proof retains each competitor's actual initial gap before summing the failure probabilities and yields the canonical Lean declarations `Tri.Multi.lemma12` and `Tri.Multi.theorem5`.

Notation

Throughout, n is the population size, x and y are the counts of the two species with $x + y = n$, $\gamma \geq 1$ is the paper's confidence parameter, and lg is the base-2 logarithm. A reaction event is *productive* if it changes the counts.

Erratum 1 (Section 4.4 — `creation-noise tolerance (P)`). Location. *Natural Computing* 19 (2020), p. 259 (Springer PDF p. 11), Sect. 4.4, first paragraph: the choice of the tolerance used to control the imbalance between the two half-rate creations.

As printed. The argument fixes the tolerance

$$d \;=\; \frac{\gamma n \lg n}{8\Delta}$$

and asserts creation-noise control at $\exp(-\Theta(\gamma \lg n))$ uniformly on the stated range $\sqrt{\gamma n \lg n}/2 < \Delta < n$.

Issue. With a per-stage creation budget of order n, the resulting Chernoff exponent is of order d^2/n, i.e. of order $\gamma^2 n \lg^2 n/\Delta^2$. This is not $\Omega(\gamma \lg n)$ uniformly on the printed range; it degrades quadratically as Δ grows and is already far too small at the LOW end of the range.

Why. At $n = 2^{20}$, $\gamma = 1$ (so $\gamma \lg n = 20$), with a budget $15n$:

Δ	d	$d^2/(2 \cdot 15n)$
$\sqrt{\gamma n \lg n}/2$	1144.9	0.042
$n/16$	40.0	1.1×10^{-4}
$n/4$	10.0	≈ 0
$n/2$	5.0	≈ 0

Every entry is far below the required $\gamma \lg n = 20$. The defect is therefore not confined to a large-Δ corner.

Fix. Use stage-specific creation budgets and deviations rather than a single global tolerance tied to Δ. The formalization does exactly that: `Tri.theorem2_singleB` carries a per-stage (D, H) pair, and the underlying concentration lemma is in fact *stronger* in temporal scope than the printed fixed-count step — `Tri/SingleBCreationTailY.lean` proves the maximal stopped bound

$$\Pr\big[\exists t:\; C_t \le H \text{ and } CY_t - CX_t \ge D\big] \le \exp\!\left(-\frac{D^2}{2H}\right)$$

with an explicit constant, where the paper states only an asymptotic $\exp(-\Theta(\gamma \lg n))$. The theorem-level conclusion of the Single-B clause is unaffected.

Erratum 2 (Theorem 4 — `preservation clause (E)`). Location. *Natural Computing* 19 (2020), pp. 261–262 (Springer PDF pp. 13–14), Theorem 4's second conclusion and its preservation argument for the subsequent n^γ interaction events.

As printed. Under $\Delta_0 := 2x - n \ge \sqrt{\gamma n \lg n}$ and $z < \Delta_0/16$, with probability $1 - \exp(-\Omega(\gamma \lg n))$ a relaxed X-consensus is reached within $O(\gamma n \lg n)$ interactions *and is preserved for the subsequent n^γ interaction events.*

Why it is false. The dynamics do not depend on the confidence parameter γ, whereas the window length n^γ does. With the Byzantine count z a fixed constant, escaping the buffer costs only $n^{-O(z)}$ per attempt, and a window of unbounded polynomial degree overwhelms it. Concretely, take

$$z = 2, \qquad \gamma = 17, \qquad (x, y, z) = (n - 16,\, 14,\, 2),$$

which lies exactly on the printed relaxed-consensus boundary $x = n - 8z$ and satisfies both hypotheses for all large n (here $\Delta_0 = n - 32$, and $(n-32)^2 \ge 17\, n \lg n$ and $2 < (n-32)/16$).

Use the paper's own declared worst response, in which Byzantine molecules always favour Y; this is a *fixed, non-adaptive* adversary, so allowing adaptive adversaries cannot rescue the claim. Partition the raw timeline into blocks of $29n^2$ interactions. From any state with $0 \le y \le 14$, one block drives the process to $y = 15$ — i.e. out of the relaxed guard — with probability at least $c\,n^{-14}$, where $c = 1/(2 \cdot 4^{14} \cdot 6^{14})$: force y down to 0; nucleate $y = 1$ from a triple $\{X, B, B\}$ whose two Byzantine molecules both present Y, so that the reaction reads $X + Y + Y \to 3Y$ and an honest X becomes an honest Y; then take fourteen prescribed up-steps, each costing at most a factor $1/(6n)$. There are $\Theta(n^{15})$ such blocks in n^{17} interactions, so

$$\Pr[\text{no exit by } n^{17}] \;\le\; \left(1 - c\,n^{-14}\right)^{\Theta(n^{15})} \;\le\; \exp(-\Theta(n)) \;\longrightarrow\; 0,$$

whereas the theorem claims an exit probability of $\exp(-\Omega(\gamma \lg n)) \to 0$.

Scale, stated honestly. The constant is small: the crossover, where $cn/29$ first exceeds 1, is near $n \approx 1.2 \times 10^{21}$. Since the theorem is asymptotic this is a legitimate refutation, and the crossover lies far below the n_0 at which this development's other headline theorems are even asserted.

Consequence. Theorem 4's *entry* conclusion is unaffected. The preservation conclusion requires either a window tied to the Byzantine count (rather than an unbounded polynomial degree), or a buffered guard with slack growing in γ. The Lean development records the printed clause in `Tri/Theorem4Statement.lean` as known-false and does not attempt it; the corrected reached-by-deadline entry clause is machine-checked as `Tri.Byzantine.theorem4_entry : Theorem4_entry_statement`, with failure at most $n^{-\gamma/136}$.

Related printed slips in the same passage. The definition of relaxed consensus by the *equality* $x = n - 8z$ cannot be preserved under any dynamics and must be read as $x \ge n - 8z$; the proof's "$x \ge n - 2z$ is equivalent to $y < z$" should read $y \le z$; and "two doublings" from $M = 2z$ reaches the boundary $M = 8z$, so strict exit needs one further upward step.

Erratum 3 (Lemma 4 — `proof`). Location. *Natural Computing* 19 (2020), p. 256 (Springer PDF p. 8), Lemma 4 proof, first displayed lower bound on the productive-reaction success probability.

As printed. After setting $d = \gamma \lg n$ and restricting to the process before the first time $y > n/k + d$, the paper prints

$$\frac{x}{n} \;\ge\; 1 - \frac{n - kd}{kn} \;>\; 1 - \frac{1}{k},$$

and concludes that each productive reaction succeeds with probability $p > 1 - 1/k$.

Issue. Both the displayed inequality and its conclusion fail. From $y \le n/k + d$ one gets $x = n - y \ge n - n/k - d$, hence

$$\frac{x}{n} \;\ge\; 1 - \frac{1}{k} - \frac{d}{n} \;=\; 1 - \frac{n + kd}{kn},$$

so the numerator is $n + kd$, not $n - kd$: the printed line overstates x/n by $2d/n$. With the correct sign the bound reads $1 - 1/k - d/n < 1 - 1/k$, so the conclusion $p > 1 - 1/k$ does not

follow, and it is genuinely false at states inside the band the proof claims to cover. The exact per-productive-event success probability

$$p_y = \frac{\binom{x}{2} y}{\binom{x}{2} y + x\binom{y}{2}} = \frac{x-1}{n-2}$$

does not repair it either. Numerical witnesses at the band's upper edge $y = n/k + d$:

k	n	d	p_y	$1 - 1/k$
4	10^3	50	0.7004	0.7500
8	10^3	50	0.8257	0.8750
6	4096	60	0.8190	0.8333
4	10^6	20	0.74998	0.75000

In each case $p_y < 1 - 1/k$, so the claimed uniform floor does not hold on the band.

Consequence. Lemma 4's conclusion is nevertheless true. The formalization (`Tri/PaperLemma4.lean`) proves it by a corrected stopped argument: with $P = n/k$ the initial minority it uses the smaller safety buffer $B = \lfloor P/8 \rfloor$, on whose band the exact productive kernel does have a uniform adaptive success floor; a Feller bound controls escape from the band and an adapted Chernoff bound controls failure to reach consensus by the paper's productive deadline. So this is a defect of the printed proof, not of the statement.

Erratum 4 (Corollary 2 — `statement`). Location. *Natural Computing* 19 (2020), p. 256 (Springer PDF p. 8), displayed statement of Corollary 2.

As printed. "Since stage s of phase 2 starts with $y \le n/2^s \le n/6$ and ends with $y = \min\{n/2^{s+1},\ \gamma \lg n\}$, it completes properly, within at most $3n/2^s$ productive reaction events, with probability at least $1 - \exp(-\Theta(\gamma \lg n))$."

Issue. The min should be a max. As printed, the stated end-value is unattainable in either regime.

Why. The paper's own definition of a phase-2 stage (L825) is unambiguous:

> "A typical stage s in phase 2 starts with $y \le n/2^s$ and continues while $n/2^{s-1} \ge y > n/2^{s+1}$. It completes properly if $y \le n/2^{s+1}$."

So the stage's own target is $y \le n/2^{s+1}$, while phase 2 as a whole terminates at its floor $y \le \gamma \lg n$ (L811). Since y is non-increasing across the relevant events, a stage that may terminate for either reason ends at whichever bound is reached *first*, i.e. at the *larger* of the two. Concretely:

- if $\gamma \lg n > n/2^{s+1}$, phase 2 has already terminated before y descends to $n/2^{s+1}$, so the printed min $= n/2^{s+1}$ is never reached within the phase;
- if $\gamma \lg n < n/2^{s+1}$, the stage completes at the dyadic threshold $n/2^{s+1}$, not at $\gamma \lg n$, so the printed min $= \gamma \lg n$ is again not the end-value.

Either way the printed min is wrong, and max is correct.

Fix. Replace min by max. Nothing downstream changes: the proof of Corollary 2 only uses that y stays above the live-band floor during the stage, which is exactly what the max form asserts.

The Lean development sidesteps the issue entirely by formalizing the *stage definition* (L825), which involves no min or max: writing $P = n/2^s$ and $B = n/2^{s+1}$, the stage runs from $y \le P$, escapes if $y > 2P$, and completes at $y \le B$, within $3P$ productive events.

Erratum 5 (Lemma 6 — `threshold`). Location. *Natural Computing* 19 (2020), p. 260 (Springer PDF p. 12), displayed statement of Lemma 6.

As printed. Under the unequal-rate hypothesis, the gap "increases to $\min\{2\Delta, n\}$" within $5n$ productive reaction events.

Issue. Classification: (T). Productive reactions change the gap by 2, and all reachable gaps have the same parity as n. Thus the displayed target cannot be read as exact equality in every population size. A concrete witness is $n = 9$, initial split $(x, y) = (5, 4)$, and $\Delta = 1$: the reachable gaps are $\{1, 3, 5, 7, 9\}$, so the literal target 2 is unreachable.

Fix. Read the conclusion as reaching the first reachable gap at least the threshold, i.e. $G_t \ge \min\{2\Delta, n\}$. This is the intended threshold statement and is what the proof supports.

Erratum 6 (Lemma 12 — `gap (P)`). Location. *Natural Computing* 19 (2020), p. 263 (Springer PDF p. 15), Lemma 12 proof, the type-(a) estimate immediately after the four stopping alternatives.

As printed (statement). "Consider a point in the computation where $x - z_{\max} = \Delta \ge \sqrt{\gamma n \lg n}$. Let $m \le n/(\gamma \lg n)$. Then $x - z_{\max} \ge \min\{2\Delta,\ n\}$ within $\Theta(n)$ productive reaction events, with probability $1 - \exp(-\Omega(\gamma \lg n))$."

As printed (the step at issue, in the proof). "By Lemma 11, completion of type (a) occurs with probability $m \exp(-\Omega(\gamma \lg n)) = \exp(-\Omega(\gamma \lg n))$ since $m < n$."

Issue. Classification: (P). The displayed absorption of the factor m into the $\Omega(\cdot)$ is not justified by $m < n$ alone. It additionally requires the constant hidden in the exponent to exceed $\ln 2$, which the paper never establishes and which the natural constants coming out of the argument do not satisfy.

Why. Write the exponent explicitly: Lemma 11 supplies a failure mass of the form $m \exp(-c\,\gamma \lg n)$ for some constant $c > 0$ determined by the argument. Using only $m \le n = 2^{\lg n}$,

$$m \exp(-c\,\gamma \lg n) \ \le\ \exp\big((\ln 2) \lg n - c\,\gamma \lg n\big) \ =\ \exp\big(-(c\gamma - \ln 2) \lg n\big).$$

For this to be $\exp(-\Omega(\gamma \lg n))$ one needs $c\gamma - \ln 2 \ge c'\gamma$ for some $c' > 0$, i.e.

$$\gamma\,(c - c') \ \ge\ \ln 2.$$

Since the lemma is asserted "for any constant $\gamma \ge 1$", this forces $c > \ln 2 \approx 0.693$. The constant that the machine-checked version of this argument actually produces is $c = 1/82944$, so for that constant the absorption fails outright for every γ in a very large range. The paper's own hypothesis $m \le n/(\gamma \lg n)$ helps only by a log factor and does not change the conclusion.

Fix. The earlier formal wrapper exposed the exact additional condition required by the printed coarse union-bound route:

$$288m + 144 \ \le\ \exp\left(\frac{\gamma \lg n}{165888}\right),$$

but the replacement proof does not use that route. For a competitor with initial gap g_Y, it retains both the Gaussian decay at the minimum gap Δ and the additional decay proportional to $g_Y - \Delta$. Competitors with population comparable to x_0 are then counted by total population rather than by m, while smaller competitors are absorbed by the additional decay. The resulting 144-substage error is at most

$$432 \exp\left(-\frac{\gamma \lg n}{663552}\right)$$

throughout the printed range $m\gamma \lg n \le n$ and $\gamma n \lg n \le \Delta^2$. This is `Tri.Multi.lemma12`.

Remark. The printed statement is true. The error is in discarding the competitor-dependent decay before summing and then trying to absorb the resulting artificial factor m.

Erratum 7 (Lemma 12 — `proof`). Location. *Natural Computing* 19 (2020), p. 264 (Springer PDF p. 16), Lemma 12 proof, the paragraph beginning "Next, consider completions of type (c)," where progress against a fixed species is followed by an invocation of Lemma 11.

As printed. The proof first obtains, for a fixed species Y, a new large X–Y gap, and then feeds that conclusion into Lemma 11.

Issue. Classification: (P). Lemma 11 assumes a *global* plurality gap $x - z_{\max} \ge \Delta'$. A large gap against one fixed Y at that species' individual hitting time does not by itself establish the global premise against every competitor. This is logically separate from the species-factor absorption error above.

Fix. The replacement proof uses one synchronized global stopping process. Its safety and progress potentials are summed over all competitors while retaining each actual initial pair gap. Consequently every terminal failure is charged once, and no fixed-Y conclusion is passed to the global Lemma 11. This proof is implemented by `productiveProperStage_progress_state`, `productiveProperStageStateError_le_exp`, and the public `lemma12`.

Erratum 8 (Lemma 12 — `terminal line`). Location. *Natural Computing* 19 (2020), p. 264 (Springer PDF p. 16), final sentence of the Lemma 12 proof.

As printed. The proof ends with the uncapped conclusion $x - z_{\max} \ge 2\Delta$.

Issue. Classification: (T). The uncapped inequality is impossible when $\Delta > n/2$. This is a terminal typesetting omission, since the lemma's own statement already uses the capped target.

Fix. The final line should read $x - z_{\max} \ge \min\{2\Delta, n\}$, with the terminal branch interpreted as consensus.

Erratum 9 (Lemma 13 — `exponent`). Location. *Natural Computing* 19 (2020), p. 264 (Springer PDF p. 16), final sentence of the Lemma 13 proof.

As printed. The proof says that λmn raw interactions, with a productive-reaction floor of order $\Theta(1/m)$, give failure $\exp(-\Theta(n/m))$.

Issue. Classification: (U). The Chernoff mean for the productive count is actually $\Theta(n)$, so the constant-fraction lower-tail calculation gives $\exp(-\Theta(n))$. The printed exponent understates the calculation.

Fix. State the stronger $\exp(-\Theta(n))$ clock tail. The weaker reading $\exp(-\Omega(n/m))$ is harmless for the downstream use.

Erratum 10 (Theorem 3 — `proof candidate`). Location. *Natural Computing* 19 (2020), p. 261 (Springer PDF p. 13), proof of Theorem 3, parameter-algebra paragraph immediately following the theorem statement.

As printed. In the proof's parameter algebra, a displayed intermediate line asserts $c < \alpha$ from assumptions including $0 < \xi < \alpha < 1$.

Issue. Classification: audit-flagged (P) candidate, not independently re-verified here. The implication $c < \alpha$ is false in general; for example, the audit reports that $\alpha = 0.1$ and $\xi = 0.05$ give $c \approx 0.859$. The line appears unnecessary for the proof, whose continuation needs $c < 1$ and the later k, β inequalities.

Fix. Delete the false intermediate inequality, or replace it by the actually used inequalities after verifying the local derivation. This entry is recorded as a candidate exactly as flagged by the audit.

Erratum 11 (Lemma 10 — `finite size`). Location. *Natural Computing* 19 (2020), p. 262 (Springer PDF p. 14), Lemma 10 proof.

As printed. From $k \geq 4$ and $x < n - 2z$, the proof derives the displayed effective-rate inequality under the paper's informal "for n sufficiently large" convention.

Issue. Classification: (P) side condition. The finite statement is only asymptotically valid. A uniform sufficient condition is

$$\gamma \lg n \leq \frac{3n}{68}.$$

The common standing assumption $6\gamma \lg n \leq n$, equivalently $\gamma \lg n \leq 11.33n/68$, is weaker and therefore does not imply this Lemma 10 instance.

Fix. Carry the strict integer form $68\gamma \lg n + 1 \leq 3n$, or an equivalent explicit hypothesis, in every formal use of Lemma 10. The corrected Theorem 4 entry declaration `Tri.Byzantine.theorem4_entry` carries exactly this side condition.

Erratum 12 (Theorem 4 — `strictness`). Location. *Natural Computing* 19 (2020), p. 262 (Springer PDF p. 14), Theorem 4 proof immediately downstream of Lemma 10.

As printed. The proof treats $x \geq n - 2z$ as implying $y < z$.

Issue. Classification: (T). Since $x + y + z = n$, the inequality $x \geq n - 2z$ is equivalent to $y \leq z$, not to $y < z$.

Fix. Replace the strict inequality by $y \leq z$ and adjust the stopping predicate accordingly.

Erratum 13 (Lemmas 17–19 — `probability qualifiers`). Location. *Natural Computing* 19 (2020), pp. 266–267 (Springer PDF pp. 18–19): the displayed statements of Lemmas 17 and 18 on p. 266, and Lemma 19 on p. 267.

As printed. The displayed lemmas have no probability qualifier.

Issue. Classification: (T), but formalization-critical. Read literally, the statements are deterministic, which is false: adverse finite reaction or activation sequences have positive probability. The proofs and downstream uses are high-probability statements.

Fix. Attach the intended high-probability qualifier to each lemma, conditional on its stated start-of-stage hypotheses and inherited epidemic events:

- Lemma 17: from the stated range $n/(\gamma n \lg n)^{1/5} \le a < n/d^*$ and $E_Y(a) \le 14c^*\sqrt{\gamma a \lg n}$, the conclusion $E_Y(2a) \le 14c^*\sqrt{\gamma(2a)\lg n}$ holds with probability $1 - \exp(-\Omega(\gamma \lg n))$.
- Lemma 18: under the preceding Lemma 17 start-of-stage bound, the stage $a = n/d^* \to 2n/d^*$ creates active X-excess at least $2Q$, where $Q = \sqrt{\gamma n \lg n}$, with probability $1 - \exp(-\Omega(\gamma \lg n))$.
- Lemma 19: if the active X-excess is at least $2Q$ at $2n/d^*$ activations, then at full activation the active X-excess is at least Q with probability $1 - \exp(-\Omega(\gamma \lg n))$.

Erratum 14 (Lemma 18 — `stopping`). Location. *Natural Computing* 19 (2020), p. 266 (Springer PDF p. 18), Lemma 18 proof, in the sentence preceding estimates (i)–(iii) and in the subsequent stopped rate estimate.

As printed. With $Q = \sqrt{\gamma n \lg n}$, the proof asserts that if the excess of active reactions of type (2) over type (1) is at most $16Q$, then the stage completes without the active Y-excess ever exceeding $30Q$.

Issue. Classification: (P), correcting our own prior classification of this point as a non-erratum. Let E_t be active Y-excess and R_t be the running reaction excess. On the preceding high-probability events, $E_0 \le 14Q$ and the activation-prefix contribution is at most Q, so

$$E_t \le 15Q + 2R_t.$$

Crossing the $30Q$ barrier therefore forces only $R_t > 15Q/2 = 7.5Q$, not $R_t > 16Q$. For example, $R_t = 10Q \le 16Q$ permits $E_t = 35Q > 30Q$. The stopping-time phrasing in step (ii) correctly makes the local rate estimate non-circular, but it does not rule out exiting through the $30Q$ barrier before the claimed endpoint reaction-excess event.

Fix. Add a maximal stopped-walk bound

$$\max_{t \le \tau} R_t \le \frac{15Q}{2}$$

with failure $\exp(-\Omega(\gamma \lg n))$, where τ is the minimum of the stage end and the first $30Q$ crossing. The paper's own rate estimate and active-reaction-count bound supply this estimate. It is stronger than the endpoint bound needed later, since $15Q/2 < 16Q$.

Erratum 15 (Heavy-B — `exact emulation claims (P)`). Location. *Natural Computing* 19 (2020), p. 258 (Springer PDF p. 10), Sect. 4.1, "Correctness of the Heavy-B emulation," last paragraph.

As printed. With $\hat{x} = x + b$ and $\hat{y} = y + b$, the paper says that "the relative probability of reaction (1') to that of (2'), equals the relative probability of reaction (1) to that of (2) in the tri-molecular CRN," and concludes that the evolution of $\hat{x}, \hat{y}$ "exactly matches" that of x, y in Tri.

Issue. The informal verb "emulate" is not itself the error. The problem is the literal pair of claims used in the Section 4.1 correctness argument: the reaction odds are asserted to be equal, and the virtual process is asserted to match Tri exactly. Neither is true in general. This is a defect in the printed proof, not a refutation of the Heavy-B clause of Theorem 2.

Why. In a Heavy-B state (x, y, b), conditioned on a resolution reaction, the favorable and adverse propensities have ratio

$$xb : yb = x : y.$$

At the virtual Tri state $(\hat{x}, \hat{y}) = (x+b, y+b)$, the two productive tri-molecular propensities have ratio

$$\binom{\hat{x}}{2}\hat{y} : \hat{x}\binom{\hat{y}}{2} = (\hat{x}-1) : (\hat{y}-1) = (x+b-1) : (y+b-1).$$

Equality would require

$$x(y+b-1) = y(x+b-1),$$

or equivalently

$$(b-1)(x-y) = 0.$$

Thus the odds agree only when $b = 1$ or $x = y$. For example, $(x, y, b) = (3, 1, 3)$ gives Heavy-B odds $3 : 1$, whereas the projected Tri odds are $5 : 3$. The raw interaction laws also use different entity populations: Heavy-B samples from $x + y + b$ entities, while the virtual population has $x + y + 2b$ entities. These two obstructions are machine-checked in `Tri/HeavyBEmulation.lean`: `Tri.heavy_tri_odds_agree_iff` proves the exact equality criterion, and `Tri.heavyToDouble_entity_count` proves the entity-count mismatch whenever $b > 0$.

Fix. Replace the exact-match argument by the one-sided comparison actually needed in the majority region: if $x \geq y$ and $b \geq 1$, the Heavy-B resolution skeleton is at least as biased toward X as productive Tri, with equality only in the two degenerate cases above. The proof must then combine this stochastic bias comparison with a Heavy-B-specific resolution clock; it cannot be obtained by black-box transport along an exact Markov lumping. The Heavy-B theorem itself is proved directly as `Tri.theorem2_heavyB`.

Erratum 16 (Single-B — `creation-imbalance concentration (P)`). Location. *Natural Computing* 19 (2020), p. 259 (Springer PDF p. 11), Sect. 4.4, Single-B phase-1 analysis, paragraph beginning "The probability that reaction $(0'_x)$ occurs is the same as the probability that reaction $(0'_y)$ occurs."

As printed. The paper observes that the two creation reactions $(0'_x)$ and $(0'_y)$ are equally likely and applies Lemma 2 to the number of $(0'_x)$ reactions among kn creation reactions. It then uses the resulting endpoint count to conclude that the virtual gap $\hat{x} - \hat{y}$ does not *become* smaller than the stated lower barrier during the stage.

Issue. Lemma 2 supplies concentration at a fixed endpoint, whereas the needed assertion is maximal: the creation-orientation imbalance must remain inside a barrier at every prefix before the stage terminates. An endpoint bound does not exclude a trajectory that crosses the adverse barrier and later returns. The printed argument also does not justify treating the history-dependent sequence of creation times as an independent sequence of trials.

Why. Let C_X and C_Y count the two creation orientations. Their masses are equal at every physical state, so $C_X - C_Y$ is an adapted fair walk when indexed by creation events. Fairness alone does not convert a terminal estimate into an ever-crossing estimate. The appropriate exponential process on the full seven-event chain is

$$G_t = \exp\left(\lambda(C_X - C_Y) - \frac{\lambda^2}{2}(C_X + C_Y)\right).$$

Inert and resolution reactions leave G_t unchanged. The two creation reactions multiply it by $\exp(\lambda - \lambda^2/2)$ and $\exp(-\lambda - \lambda^2/2)$ with equal mass, and

$$\frac{e^{\lambda} + e^{-\lambda}}{2} \le e^{\lambda^2/2}.$$

Hence G_t is a supermartingale without an independence assumption. If the number of creations is at most H, Ville's inequality gives

$$\Pr[\exists t:\ C_X(t) - C_Y(t) \ge D,\ C_X(t) + C_Y(t) \le H] \le \exp\left(-\frac{D^2}{2H}\right),$$

and the sign-reversed potential gives the opposite tail.

Fix. Replace the endpoint invocation of Lemma 2 by the stopped maximal estimate above, retain an explicit creation-count budget, and combine both creation tails with the corrected resolution-gap process before recovering the physical gap. The formalization proves the equal-mass input as `Tri.singleCompPMF_creation_masses_eq`, the full-kernel one-step supermartingale as `Tri.creationG_step`, the two maximal bounds as `Tri.creation_tail_maximal` and `Tri.creationY_tail_maximal`, and their optimized forms as `Tri.creation_tail` and `Tri.singleCreationY_tail`. The complete Single-B clause is `Tri.theorem2_singleB`.

Erratum 17 (Lemma 8 — `adaptive-policy comparison (P)`). Location. *Natural Computing* 19 (2020), p. 261 (Springer PDF p. 13), Lemma 8 and its proof in the Byzantine section.

As printed. For $0 < \ell \le n - x - z$, Lemma 8 says that

$$\pi^*((x, y, z), (x + \ell, y - \ell, z))$$

is minimized by the paper's fixed Byzantine response. The proof says it suffices to observe

$$\pi^*((x, y, z), q) \le \pi^*((x + 1, y - 1, z), q), \qquad q = (x + \ell, y - \ell, z),$$

because every path from the first state to q passes through the second.

Issue. The displayed inequality compares two starting states under one fixed transition law. The lemma's conclusion compares different Byzantine response policies, including responses that may depend on the complete past. Starting one step closer to a target under a fixed policy does not establish that one policy minimizes the target-hitting probability.

Why. Skip-freeness proves only the stated spatial monotonicity: for a fixed policy, every successful path from x to $x+\ell$ first visits $x+1$. A policy comparison needs a Bellman argument. For each finite horizon, the continuation value—the probability of reaching the upper target in the remaining steps—must first be proved monotone in the current X count. The pointwise worst-response inequality can then be applied to this monotone continuation value, followed by backward induction over the horizon and over the complete history. The eventual hitting comparison is obtained by taking the increasing supremum over finite horizons. Those policy-comparison steps are absent from the printed proof.

Fix. Replace the proof by finite-horizon dynamic programming and then pass to the unbounded hitting event. This repair is machine-checked as `Tri.Byzantine.lemma8`. Its finite-horizon core `lemma8_finite_horizon` proves Bellman domination against every history-dependent strategy; nearest-neighbour support identifies the upper Bellman target with the exact configuration, and a supremum over all finite horizons gives the paper's unbounded π^* comparison.

Erratum 18 (Lemma 11 — `ratio inequalities (P)`). Location. *Natural Computing* 19 (2020), p. 263 (Springer PDF p. 15), Lemma 11 proof, the two-party and third-party backsliding calculations.

As printed. In the two-party calculation the paper sets $p = 1/2 + \Delta/(4n)$ and prints

$$\left(\frac{1-p}{p}\right)^{\Delta/8} < \left(1 - \frac{\Delta}{n}\right)^{\Delta/8}.$$

In the third-party calculation it sets $p = 1/2 + \Delta/(8n)$ and prints

$$\left(\frac{1-p}{p}\right)^{\Delta/4} < \left(1 - \frac{\Delta}{2n}\right)^{\Delta/4}.$$

Issue. Both displayed comparisons are false: before taking powers, the left-hand ratio is strictly *larger* than the printed linear factor. These two displayed comparisons are the entirety of the "false ratio inequalities" defect. No stronger defect in Lemma 11 is asserted: the paper's two-mode decomposition is legitimate, and its stated $m \exp(-\Omega(\Delta^2/n))$ conclusion survives after repairing these two estimates.

Why. In the two-party case,

$$\frac{1-p}{p} = \frac{2n-\Delta}{2n+\Delta},$$

and

$$\frac{2n-\Delta}{2n+\Delta} - \left(1 - \frac{\Delta}{n}\right) = \frac{\Delta^2}{n(2n+\Delta)} > 0.$$

For example, $n = 100$, $\Delta = 10$ gives $190/210 \approx 0.90476 > 0.9$. In the third-party case,

$$\frac{1-p}{p} = \frac{4n-\Delta}{4n+\Delta},$$

and

$$\frac{4n-\Delta}{4n+\Delta} - \left(1 - \frac{\Delta}{2n}\right) = \frac{\Delta^2}{2n(4n+\Delta)} > 0.$$

For $n = 100$, $\Delta = 10$ this is $390/410 \approx 0.95122 > 0.95$.

Fix. Use the valid inequality

$$\frac{1-t}{1+t} \le e^{-2t} \qquad (0 \le t < 1).$$

With $t = \Delta/(2n)$ in the two-party case,

$$\left(\frac{1-p}{p}\right)^{\Delta/8} \le \exp\left(-\frac{\Delta^2}{8n}\right).$$

With $t = \Delta/(4n)$ in the third-party case,

$$\left(\frac{1-p}{p}\right)^{\Delta/4} \le \exp\left(-\frac{\Delta^2}{8n}\right).$$

Union over the two modes and the $m-1$ competitors therefore preserves the paper's asymptotic form $m\exp(-\Omega(\Delta^2/n))$. The Lean development avoids the two erroneous linear comparisons and proves the all-time stopped estimate directly by a geometric supermartingale. The core theorem is `Tri.Multi.productivePairGapEver_half_failure_exp`; its two paper-facing wrappers are `Tri.Multi.lemma11_twoPartyBacksliding` and `Tri.Multi.lemma11_thirdPartyBacksliding` in `Tri/PaperLemma11.lean`, with the explicit envelope

$$m\exp\left(-\frac{\Delta^2}{18n}\right).$$

Non-errata: differences that are ours, not the paper's

For the record, the following divergences between the Lean development and the paper are *our* choices or *our* earlier mistakes, and are deliberately not listed above.

- Phase 0 of Theorem 1(a) is proved with a concave symmetric Lyapunov supermartingale rather than the paper's foray/checkpoint combinatorics. Both are correct; ours was easier to formalize.
- Corollary 3's proof omits a monotonicity step: the productive-clock estimate may be proved at the boundary minority size $y = \gamma \lg n$ and then pushed to starts with smaller minority. This is a true omitted step, not an erratum, because the statement remains correct. The formalization records the missing comparison as `Tri.corollary3_boundary_success_le_entry_success`, built from the reusable Bellman comparison `Tri.productive_hitProb_mono_start_of_upper`.
- An early version of our Corollary 3 wrapper stated a raw-interaction result under the productive-clock name. That was our labelling error, now corrected, and is not a defect of the paper.
- Lemma 18's deterministic endpoint ledger closes with *zero* slack. The paper's own arithmetic is $48\sqrt{\gamma n \lg n} - 14\sqrt{\gamma n \lg n} - 32\sqrt{\gamma n \lg n} = 2\sqrt{\gamma n \lg n}$, exactly the required gap. A formalization has no room to lose a constant anywhere in this lemma.